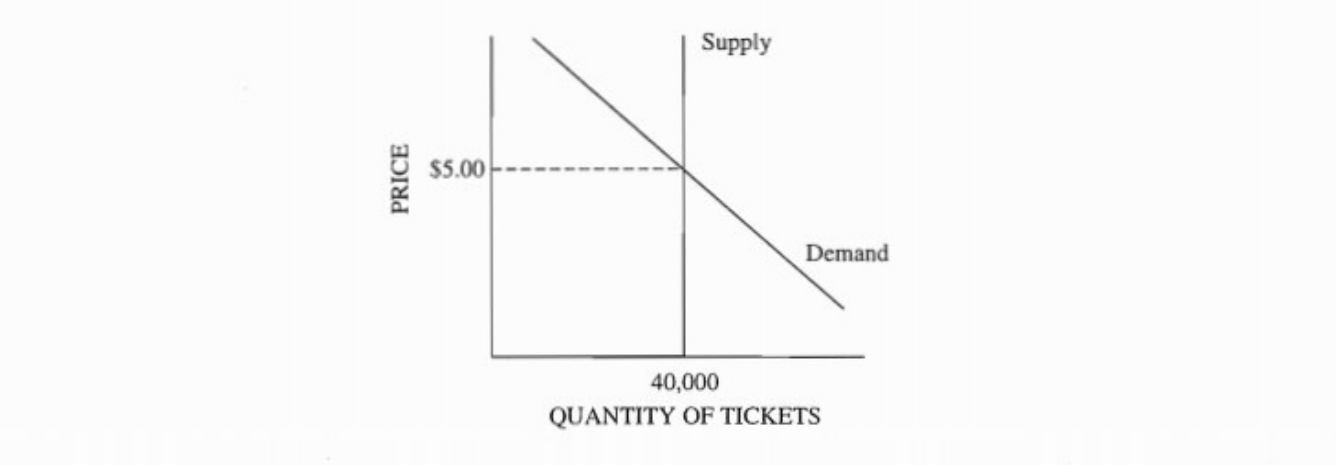
2.8 government intervention FRQ



The Toledo arena holds a maximum of 40,000 people, as indicated in the graph above. Each year the circus holds eight performances, all of which are sold out.

a. Analyze the effect on each of the following of the addition of a fantastic new death-defying trapeze act that increases the demand for tickets.

   i. The price of tickets

   ii. The quantity of tickets sold

b. The city of Toledo institutes an effective price ceiling on tickets. Explain where the price ceiling would be set. Explain the impact of the ceiling on each of each of the following.

   i. The quantity of tickets demanded

   ii. The quantity of tickets supplied

c. Will everyone who attends the circus pay the ceiling price set by the city of Toledo? Why or why not?

1. Respond to all parts of the question.

Part A

2 Points

1 - Price increases as a result of the demand increase

1 - Quantity remains the same

0	1	2 ✓

2.8 government intervention FRQ

The student earns all of the following points:

1 - Price increases as a result of the demand increase

1 - Quantity remains the same

Part B

2 Points

- **1 -** Setting the price ceiling below the equilibrium price.
- **1 -** Quantity demanded increases
- **1/2 -** Quantity supplied unchanged

0	1	2 ✓

The student earns all of the following points:

- **1 -** Setting the price ceiling below the equilibrium price.
- **1 -** Quantity demanded increases
- **1/2 -** Quantity supplied unchanged

Part C

1 Point

1 - for no, a secondary market or reselling of tickets or scalping of tickets will have some circus-goers paying more than the ceiling price.

0	1 ✓

The student earns all of the following points:

- **1 -** for no, a secondary market or reselling of tickets or scalping of tickets will have some circus-goers paying more than the ceiling price.

2.8 government intervention FRQ

Assume that bread and butter are complementary goods. The government begins to subsidize the production of wheat, which is an input in the production of bread.

a. For each of the following markets, draw correctly labeled supply and demand graphs and show the effect of the subsidy on the equilibrium price and quantity in the short run.

   i. The wheat market

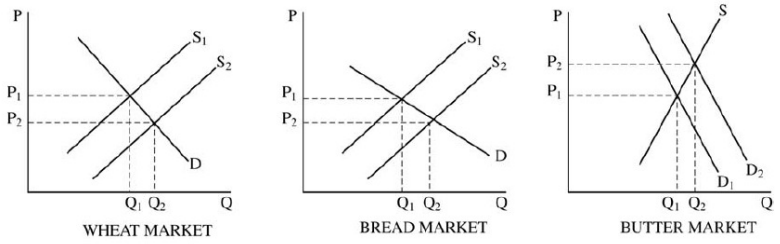
   ii. The bread market

   iii. The butter market

b. If the demand for bread is price elastic, how will total revenues for the bread producers change as a result of the government subsidy?

2. Respond to all parts of the question.

Part A



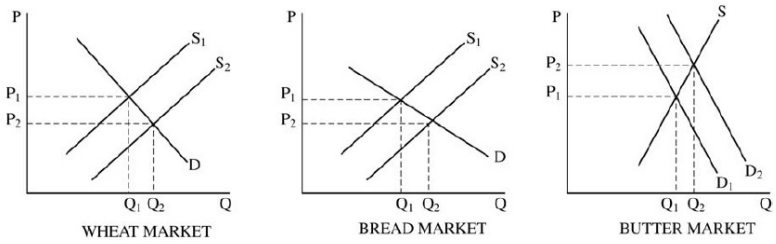
4 points:

- One point is earned for correctly labeled supply and demand graphs for all three markets.
- One point is earned for showing a rightward shift of the supply curve in the wheat market with price decreasing and quantity increasing.
- One point is earned for showing a rightward shift of the supply curve in the bread market with price decreasing and quantity increasing.
- One point is earned for showing a rightward shift of the demand curve in the butter market with price increasing and quantity increasing.

0	1	2	3	4 ✓

Student receives 4/4 point(s)

2.8 government intervention FRQ



4 points:

- One point is earned for correctly labeled supply and demand graphs for all three markets.
- One point is earned for showing a rightward shift of the supply curve in the wheat market with price decreasing and quantity increasing.
- One point is earned for showing a rightward shift of the supply curve in the bread market with price decreasing and quantity increasing.
- One point is earned for showing a rightward shift of the demand curve in the butter market with price increasing and quantity increasing.

Part B

1 point:

- One point is earned for concluding that the total revenue will increase.

0	1 ✓

Student receives 1/1 point(s)

1 point:

- One point is earned for concluding that the total revenue will increase.

2.8 government intervention FRQ

Assume that the market for home security systems is perfectly competitive and currently in equilibrium.

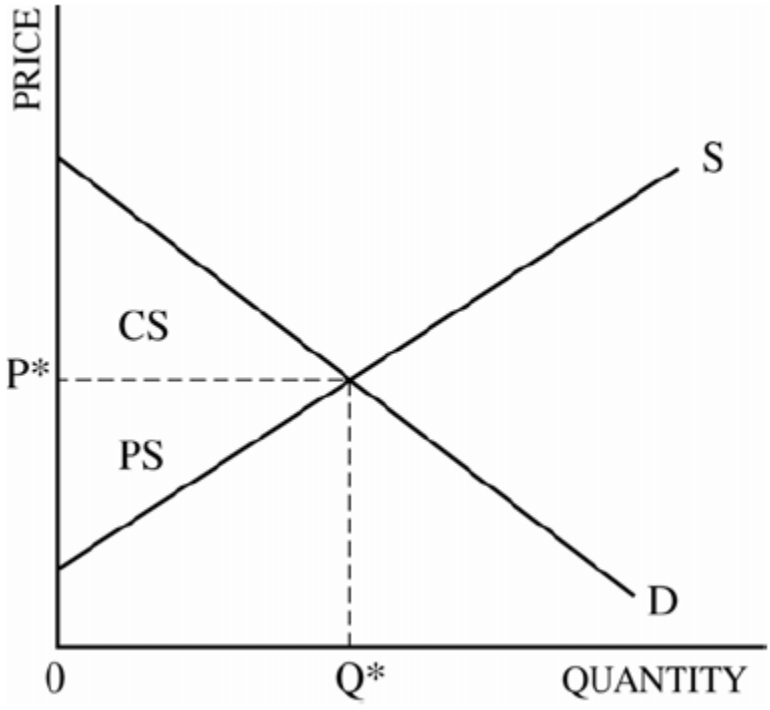
a. Using a correctly labeled graph of supply and demand, show each of the following.
    i. The equilibrium price and quantity, labeled as P* and Q*, respectively
    ii. The area representing consumer surplus, labeled as CS
    iii. The area representing producer surplus, labeled as PS

b. Suppose that the government imposes an effective (binding) price ceiling. Redraw your graph in part (a), and label the ceiling price as P_2. Completely shade the area representing the sum of the consumer surplus and the producer surplus after the imposition of the price ceiling.

c. Suppose the demand for home security systems decreases and the price ceiling remains binding. Indicate what will happen to each of the following.
    i. Consumer surplus
    ii. Producer surplus

3. Respond to all parts of the question.

Part A



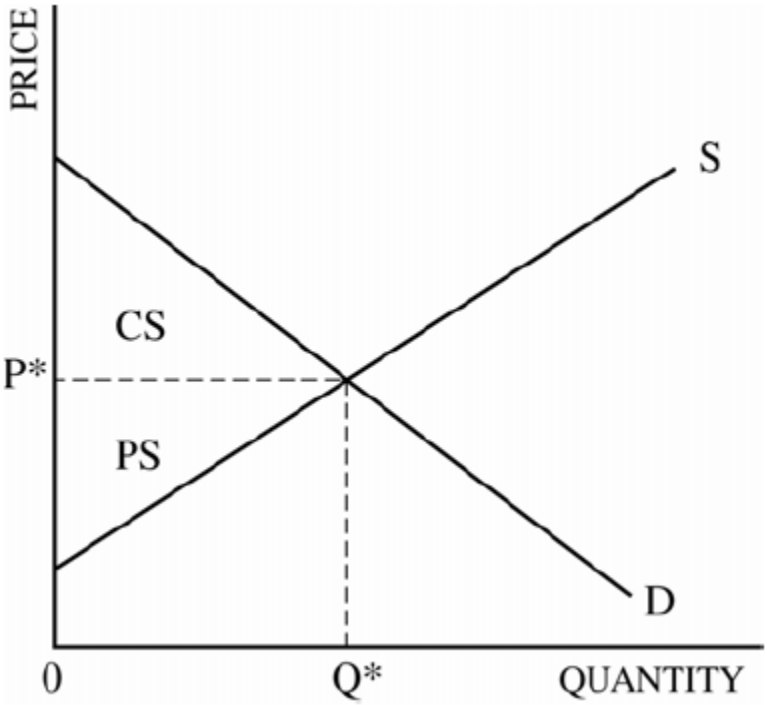
3 points:

2.8 government intervention FRQ

- One point is earned for a correctly labeled graph with equilibrium price and quantity.
- One point is earned for showing the correct area of consumer surplus.
- One point is earned for showing the correct area of producer surplus.

0	1	2	3 ✓

Student receives 3/3 point(s)

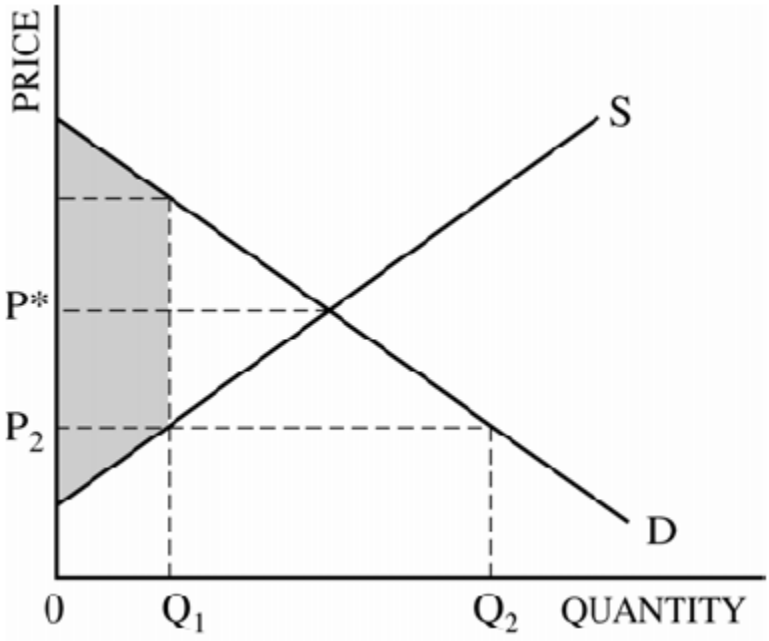


3 points:

- One point is earned for a correctly labeled graph with equilibrium price and quantity.
- One point is earned for showing the correct area of consumer surplus.
- One point is earned for showing the correct area of producer surplus.

Part B

2.8 government intervention FRQ



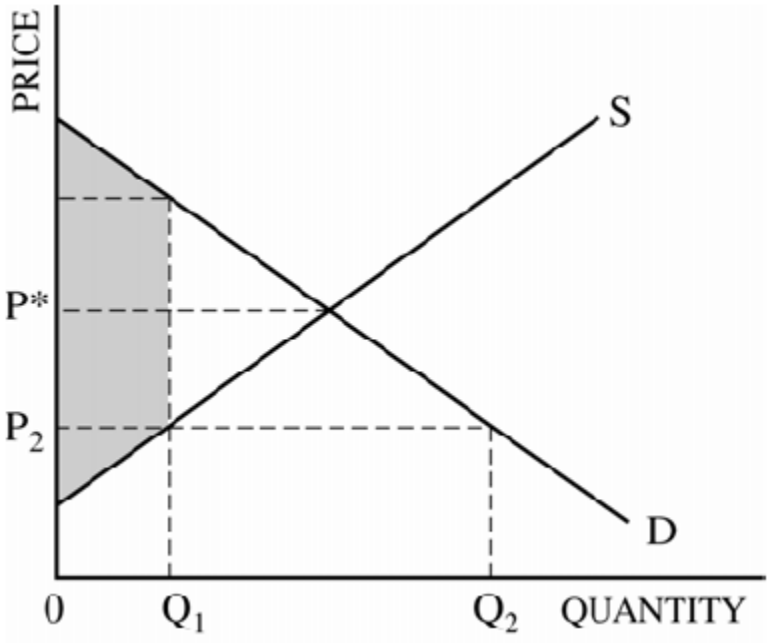
2 points:

- One point is earned for showing the price ceiling below the equilibrium.
- One point is earned for shading the correct area of consumer surplus and producer surplus.

0	1	2 ✓

Student receives 2/2 point(s)

2.8 government intervention FRQ



2 points:

- One point is earned for showing the price ceiling below the equilibrium.
- One point is earned for shading the correct area of consumer surplus and producer surplus.

Part C

2 points:

- One point is earned for indicating that consumer surplus will decrease.
- One point is earned for indicating that producer surplus will not change.

0	1	2 ✓

Student receives 2/2 point(s)

2 points:

- One point is earned for indicating that consumer surplus will decrease.
- One point is earned for indicating that producer surplus will not change.

2.8 government intervention FRQ

Directions: You are advised to spend the first 10 minutes reading all of the questions and planning your answers. You will then have 50 minutes to answer all three of the following questions. You may begin writing your responses before the reading period is over. It is suggested that you spend approximately half your time on the first question and divide the remaining time equally between the next two questions. Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. Use a pen with black or dark blue ink.

The markets for bananas, muffins, and coffee are interrelated, and each market is perfectly competitive.

a. *In the market for bananas, the equilibrium price is $1.00 per pound, and the equilibrium quantity is 1,000 pounds per week. Suppose the government imposes a price floor on bananas at $1.20* per pound, causing the quantity supplied to increase to 1,500 pounds per week.

   i. Would the price floor result in a shortage, a surplus, or neither? Explain.

   ii. Calculate the price elasticity of supply if the price increases from $1 to $1.20. Show your work.

   iii. Between $1 and $1.20, is the supply elastic, unit elastic, or inelastic? Explain.

b. Bananas are an input for muffins.

   i. Draw a correctly labeled graph of the market for muffins indicating the equilibrium price and quantity, labeled P_0 and Q_0, respectively.

   ii. On the graph drawn in part (b)(i), show the impact of an increase in the price of bananas on the muffin market, labeling the new equilibrium price and quantity P_1 and Q_1, respectively.

   iii. On the same graph, completely shade the area that represents the **change** in the consumer surplus caused by the increase in the price of bananas.

c. In the market for coffee, the equilibrium price is $3.00 per cup and the equilibrium quantity is 100 cups per week. The cross-price elasticity of coffee with respect to muffins is -2.

   i. Are coffee and muffins normal goods, inferior goods, complementary goods, or substitute goods?

   ii. Assume the supply of coffee is perfectly elastic. Using the equilibrium price and quantity given above, draw a correctly labeled graph for the coffee market, and show the impact of an increase in the price of muffins on the coffee market.

   iii. Given the original quantity of 100 cups of coffee per week, if the increase in the price of muffins is 10%, calculate the new equilibrium quantity in the coffee market. Show your work.

4. Respond to all parts of the question.

Part A

2.8 government intervention FRQ

1 point is earned for: stating that the quantity supplied exceeds the quantity demanded at the price floor **or** the price floor would result in a surplus because the price floor is binding or effective.

1 point is earned for: correctly calculating the price elasticity of supply.Students can use either the mid-point formula or the point elasticity formula.

E = (500/1250)/ (0.2/1.1) = 2.2

E = (50%/20%)= 2.5E= (500/0.2) ×(1/1000) = 2.5

E = [(1500–1000)/1000]/[(1.2–1)/1]= 0.5/0.2 = 2.5

E = (500/0.2) x (1.2/1500) = 2

1 point is earned for: stating that the supply is elastic and for explaining that the **percentage** change in quantity supplied exceeds the **percentage** change in price or because price elasticity of supply is greater than one. (Other equivalent explanations are accepted.)

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for: stating that the quantity supplied exceeds the quantity demanded at the price floor **or** the price floor would result in a surplus because the price floor is binding or effective.

1 point is earned for: correctly calculating the price elasticity of supply.Students can use either the mid-point formula or the point elasticity formula.

E = (500/1250)/ (0.2/1.1) = 2.2

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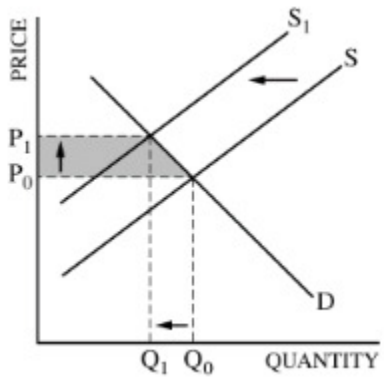
E = [(1500–1000)/1000]/[(1.2–1)/1]= 0.5/0.2 = 2.5

E = (500/0.2) x (1.2/1500) = 2

1 point is earned for: stating that the supply is elastic and for explaining that the **percentage** change in quantity supplied exceeds the **percentage** change in price or because price elasticity of supply is greater than one. (Other equivalent explanations are accepted.)

Part B

2.8 government intervention FRQ



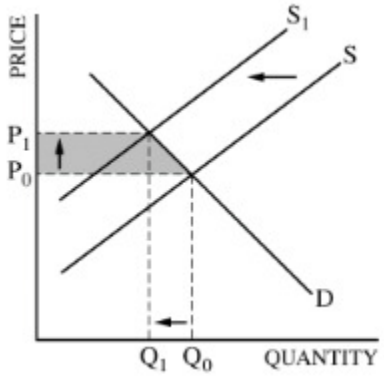
1 point is earned for: drawing a correctly labeled graph and for showing the equilibrium price and quantity, labeled P0 and Q0, respectively.

1 point is earned for: shifting the supply curve to the left and for showing the new equilibrium price and quantity, labeled P1 and Q1, respectively.

1 point is earned for: completely shading on the graph the area representing the change in consumer surplus.

0	1	2	3 ✓

The student response earns all of the following points:



1 point is earned for: drawing a correctly labeled graph and for showing the equilibrium price and quantity, labeled P0 and Q0, respectively.

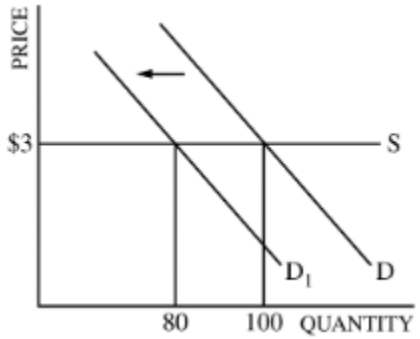
1 point is earned for: shifting the supply curve to the left and for showing the new equilibrium price and quantity, labeled P1 and Q1, respectively.

1 point is earned for: completely shading on the graph the area representing the change in consumer surplus.

Part C

1 point is earned for: stating that coffee and muffins are complementary goods.

2.8 government intervention FRQ



1 point is earned for: drawing a correctly labeled graph for the coffee market and showing a horizontal supply curve(S)and a downward sloping demand curve(D).

1 point is earned for: shifting the demand curve to the left and for showing a decrease in the equilibrium quantity and no change in the equilibrium price of coffee.

1 point is earned for: correctly calculating the new equilibrium quantity and showing the work.

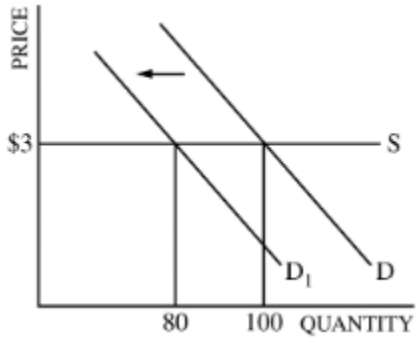
$\%\Delta Q = (10\% \times -2) = -20\%$
New Equilibrium Quantity = $100 \times (-20\%) = 80$
OR
New Equilibrium Quantity = $100 - (0.2 \times 100) = 80$

(Using the midpoint formula is also acceptable.)

0	1	2	3	4 ✓

The student response earns all of the following points:

1 point is earned for: stating that coffee and muffins are complementary goods.



1 point is earned for: drawing a correctly labeled graph for the coffee market and showing a horizontal supply curve(S)and a downward sloping demand curve(D).

1 point is earned for: shifting the demand curve to the left and for showing a decrease in the equilibrium quantity and no

2.8 government intervention FRQ

change in the equilibrium price of coffee.

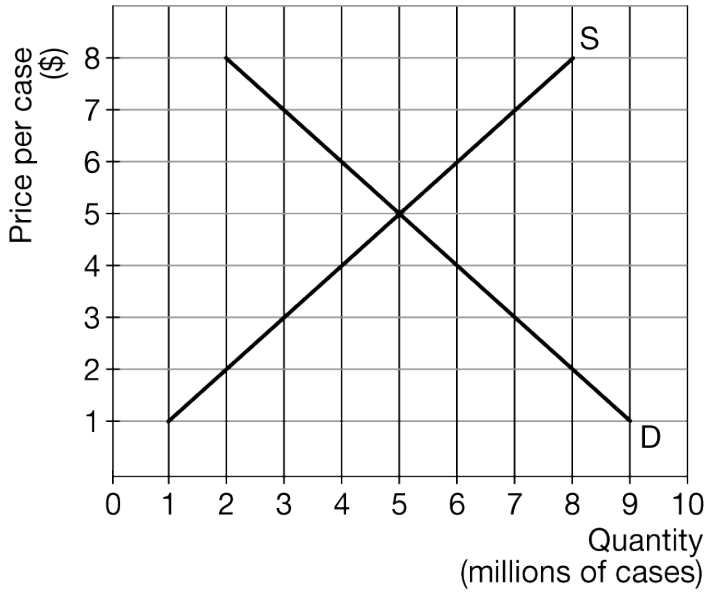
1 point is earned for: correctly calculating the new equilibrium quantity and showing the work.

$\%\Delta Q = (10\% \times -2) = -20\%$
New Equilibrium Quantity = $100 \times (-20\%) = 80$
OR
New Equilibrium Quantity = $100 - (0.2 \times 100) = 80$

(Using the midpoint formula is also acceptable.)

2.8 government intervention FRQ

Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.



Assume that sugar-based soft drinks are produced in a market shown on the graph above. Answer the following questions based on the information given in the graph.

(a) To reduce the consumption of sugary soft drinks, suppose the government imposes a $2 per-unit sales tax on soft drinks.

(i) Will the price of soft drinks increase by the full amount of the per-unit sales tax? Explain.

(ii) Calculate the tax revenue the government can collect from the sale of soft drinks. Show your work.

(iii) Will the consumer surplus increase, decrease, or stay the same after the per-unit sales tax?

(iv) Calculate the deadweight loss created by the per-unit sales tax. Show your work.

(b) Suppose that instead of imposing the per-unit sales tax, the government sets a price ceiling of $7. Identify the quantity of soft drinks that will be exchanged in the market as a result of the price ceiling. Explain.

5. Respond to all parts of the question.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

All four criteria in part (a) are achieved independently.

2.8 government intervention FRQ

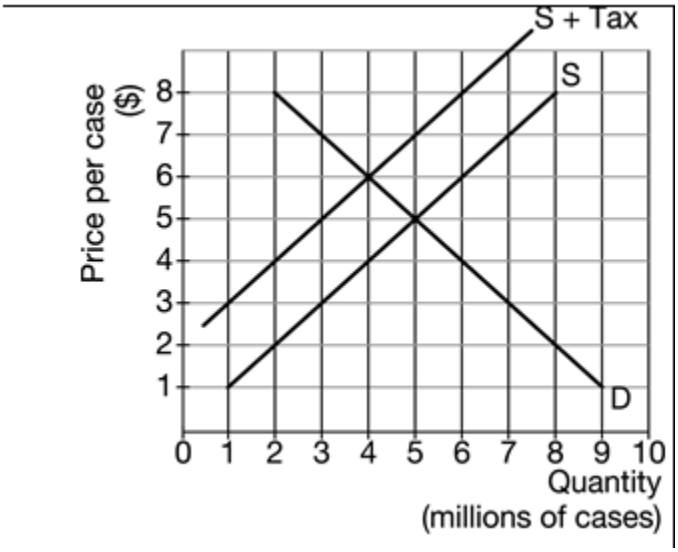
0	1	2	3	4 ✓

The response accurately includes all four of the criteria below.

- ☐ States the price will not increase by the full amount of the per-unit sales tax and explains that the per-unit sales tax will shift the supply curve to the left (or upward) by the full amount of the tax, but the equilibrium price with the tax will only rise from \$5 to \$6.
- ☐ Calculates the tax revenue as \$8 million and shows appropriate work as : $(T \times Q) = (\$2 \times 4 \text{ million}) = \$8 \text{ million}$
- ☐ States that the consumer surplus will decrease.
- ☐ Calculates the deadweight loss as \$1 million and shows appropriate work as : $\frac{1}{2}(\$6 - \$4)(1 \text{ million}) = \$1 \text{ million}$

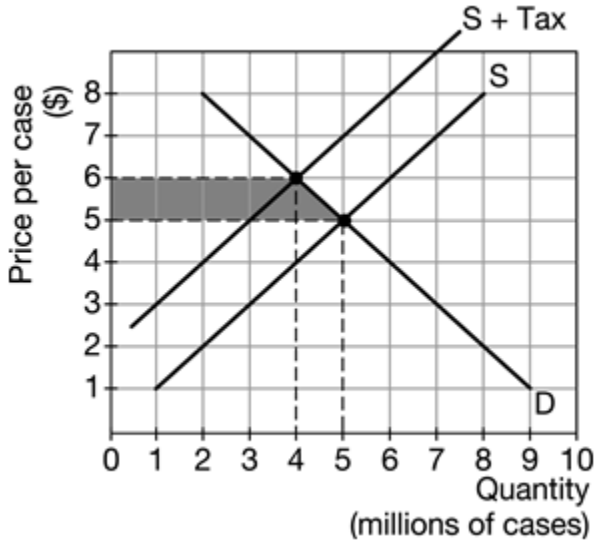
Supporting graphs (for reference only; not required for a correct response):

The graph below demonstrates the effect of the per-unit sales tax on the new equilibrium price of soft drinks.

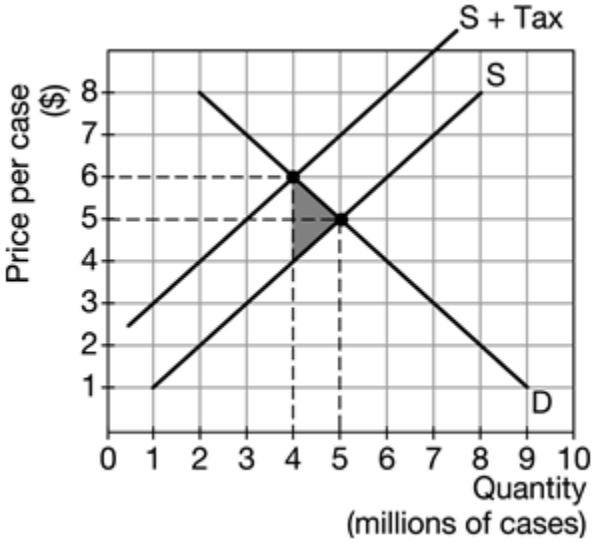


The graph below illustrates the area of consumer surplus before and after the tax is imposed.

2.8 government intervention FRQ



The graph below illustrates the area of deadweight loss.



Part (b)

elect a point value to view scoring criteria, solutions, and/or examples and to score the response.

General Considerations

Both the identification of the quantity as 5 million cases of soft drinks and the explanation are required to receive the 1 point.

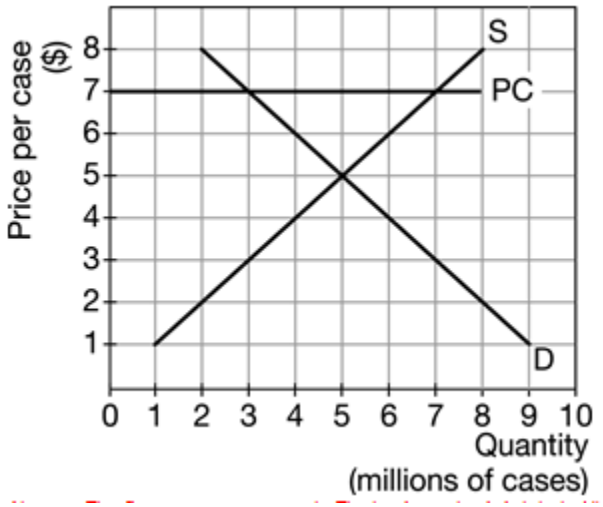
The response should recognize that the price ceiling is not effective or binding because the price ceiling is set above the equilibrium price.

2.8 government intervention FRQ

0	1 ✓

The response states that the quantity exchanged in the market is 5 million cases of soft drinks (see graph below) and explains that the price ceiling is not effective or binding because the price ceiling is set above the equilibrium price.

(The graph below illustrating the price ceiling is provided as reference and may be used as part of the explanation but is not required.)



In each part below, assume that the government imposes a per-unit sales tax and that the supply curve is upward-sloping.

a. In industry X, consumers buy the same quantity no matter what the price is.

   i. Using a correctly labeled graph, show what happens to the quantity sold when the tax is imposed.

   ii. How will the burden of the tax be distributed between buyers and sellers?

b. In industry Y, the market demand curve is perfectly elastic.

   i. Using a correctly labeled graph, show what happens to the price of the good that the consumers pay when the tax is imposed.

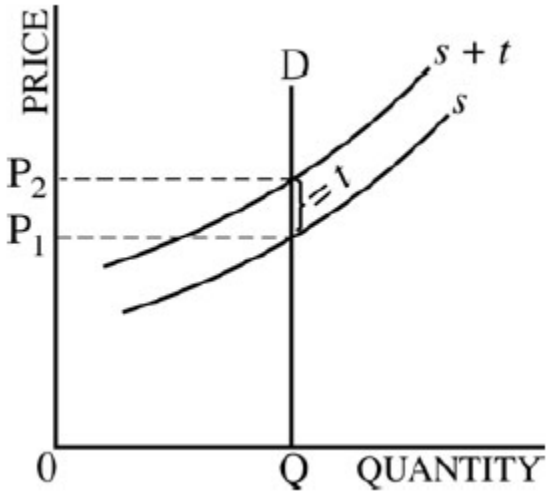
   ii. How will the burden of the tax be distributed between buyers and sellers?

c. In industry Z, the market demand curve is downward-sloping. Using a correctly labeled graph, shade the area that represents total tax revenues.

2.8 government intervention FRQ

6. Respond to all parts of the question.

Part A

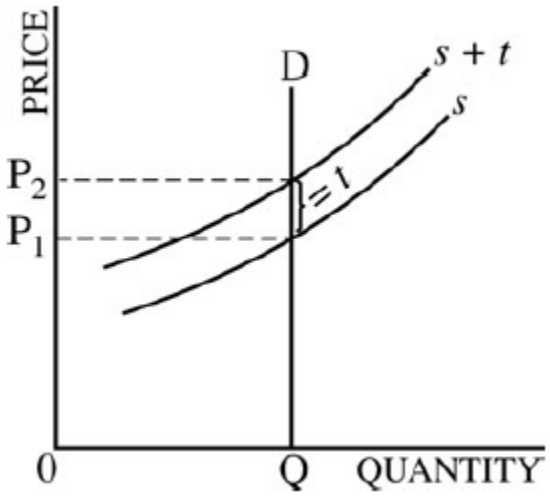


1 point - Perfectly inelastic demand curve showing that Q does not change

1 point - Since producers can raise the price by the full amount of the tax, the tax falls entirely on buyers

0	1	2 ✓

Student receives 2/2 point(s)

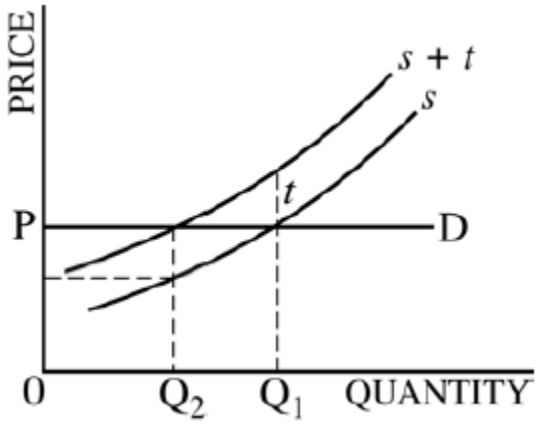


2.8 government intervention FRQ

1 point - Perfectly inelastic demand curve showing that Q does not change

1 point - Since producers can raise the price by the full amount of the tax, the tax falls entirely on buyers

Part B



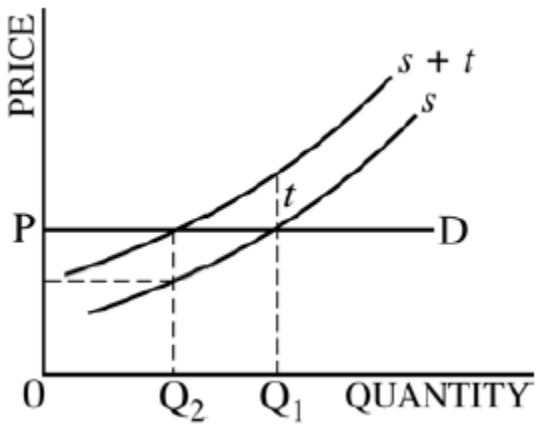
1 point – Horizontal demand curve showing that price does not change

1 point – The tax falls entirely on sellers, since they can't charge more and thus must absorb the entire amount of the tax

✓

0	1	2

Student receives 2/2 point(s)

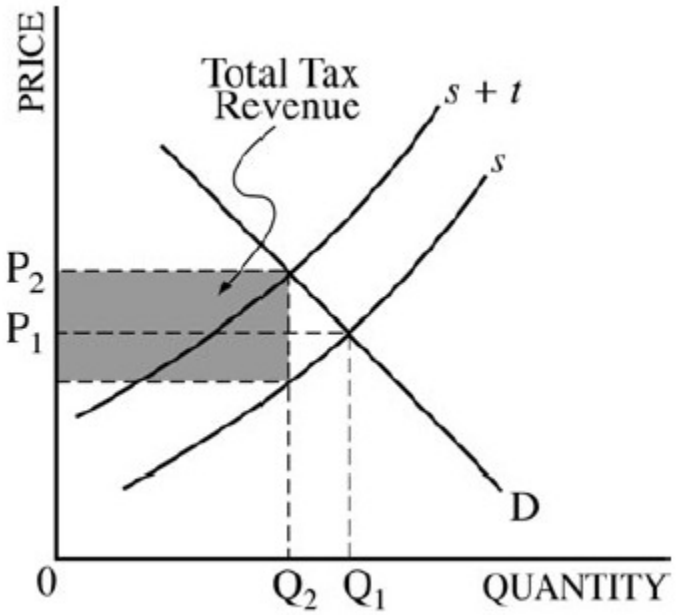


1 point – Horizontal demand curve showing that price does not change

1 point – The tax falls entirely on sellers, since they can't charge more and thus must absorb the entire amount of the tax

Part C

2.8 government intervention FRQ

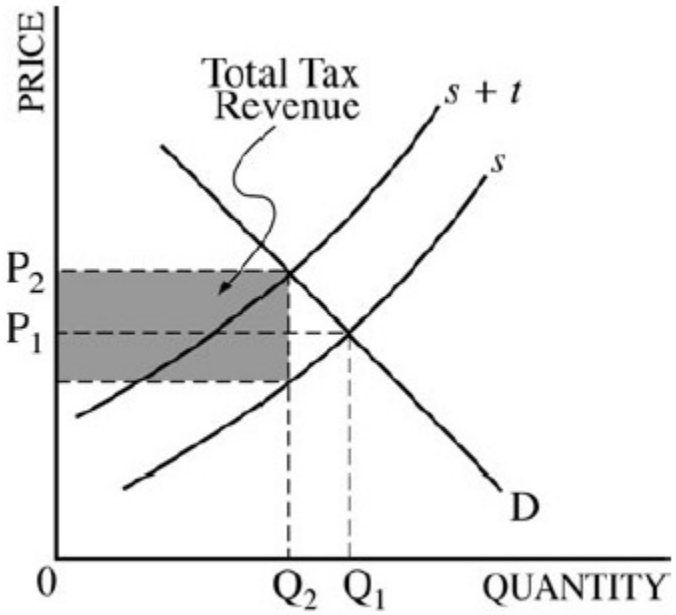


1 point - For shifting either the supply curve *or* the demand curve inward to the left

1 point - For shading the correct profits area

0	1	2 ✓

Student receives 2/2 point(s)

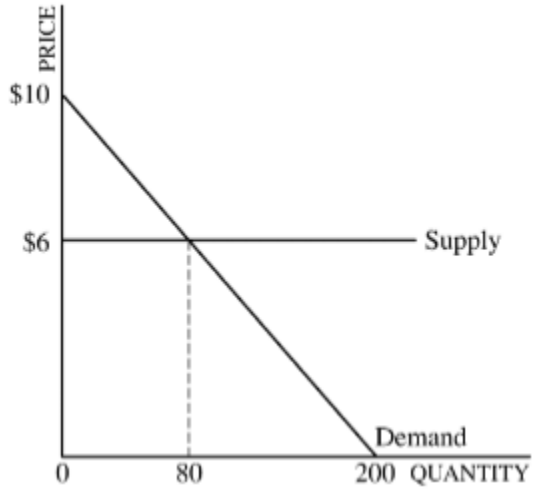


2.8 government intervention FRQ

1 point - For shifting either the supply curve *or* the demand curve inward to the left

1 point - For shading the correct profits area

The market supply and demand for a product are shown in the diagram below.



7. Respond to all parts of the question.

a. Is the price elasticity of supply less than one, equal to one, or greater than one? Explain.

b. Calculate consumer surplus at the equilibrium price. Show your work.

c. Now suppose the government imposes a per-unit tax of $1 on producers.

i. What happens to total revenue received by producers after they pay the tax to the government? Explain.

ii. Will producer surplus increase, decrease, or stay the same?

iii. Will total surplus increase, decrease, or stay the same? Explain.

Part A

1 point is earned for: stating that the price elasticity of supply is greater than one and for explaining that the supply curve is horizontal **OR** the price elasticity of supply is infinite **OR** the supply is perfectly elastic.

0	1 ✓

2.8 government intervention FRQ

The student response earns all of the following points:

1 point is earned for: stating that the price elasticity of supply is greater than one and for explaining that the supply curve is horizontal **OR** the price elasticity of supply is infinite **OR** the supply is perfectly elastic.

Part B

1 point is earned for: correctly calculating the consumer surplus and showing the work as 1/2 × 80 × $4 = $160.

0	1 ✓

The student response earns all of the following points:

1 point is earned for: correctly calculating the consumer surplus and showing the work as 1/2 × 80 × $4 = $160.

Part C

1 point is earned for: stating that the total revenue will decrease and explaining that the quantity will decrease **OR** the percentage change of quantity demanded is greater than the percentage change in price **OR** demand is elastic in this price range.

1 point is earned for: stating that producer surplus will stay the same **OR** remain zero.

1 point is earned for: stating that total surplus will decrease and for explaining that the quantity decreases **OR** there is now deadweight loss after the tax is imposed **OR** the consumer surplus decreases by more than what is transferred to the government and producer surplus stays the same.

0	1	2	3 ✓

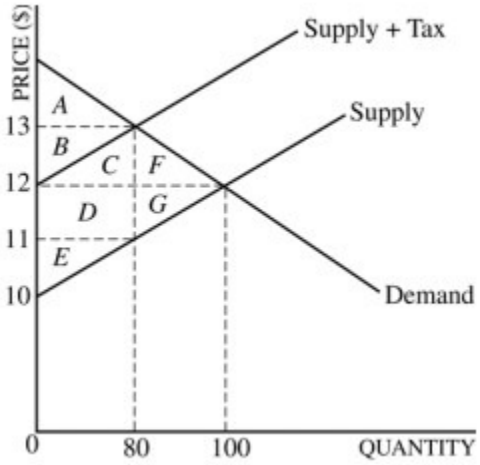
The student response earns all of the following points:

1 point is earned for: stating that the total revenue will decrease and explaining that the quantity will decrease **OR** the percentage change of quantity demanded is greater than the percentage change in price **OR** demand is elastic in this price range.

1 point is earned for: stating that producer surplus will stay the same **OR** remain zero.

1 point is earned for: stating that total surplus will decrease and for explaining that the quantity decreases **OR** there is now deadweight loss after the tax is imposed **OR** the consumer surplus decreases by more than what is transferred to the government and producer surplus stays the same.

2.8 government intervention FRQ



The graph above shows the market for a good that is subject to a per-unit tax. The letters in the graph represent the enclosed areas.

a. Using the labeling on the graph, identify each of the following.

   i. The equilibrium price and quantity before the tax

   ii. The area representing the consumer surplus before the tax

   iii. The area representing the producer surplus before the tax

b. Assume that the tax is now imposed. Based on the graph, does the price paid by the buyers rise by the full amount of the tax? Explain.

c. Using the labeling on the graph, identify each of the following after the imposition of the tax.

   i. The net price received by the sellers

   ii. The amount of tax revenue

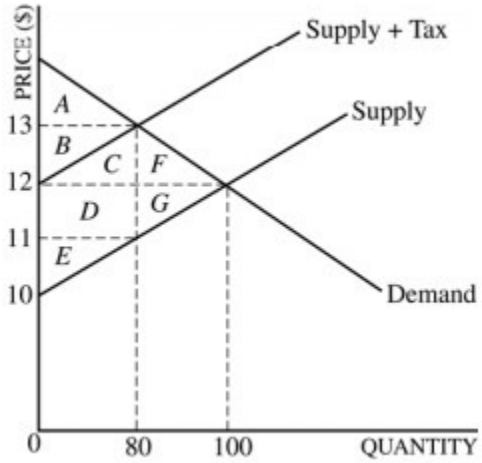
   iii. The area representing the consumer surplus

   iv. The area representing the deadweight loss

8. Respond to all parts of the question.

Part A

2.8 government intervention FRQ

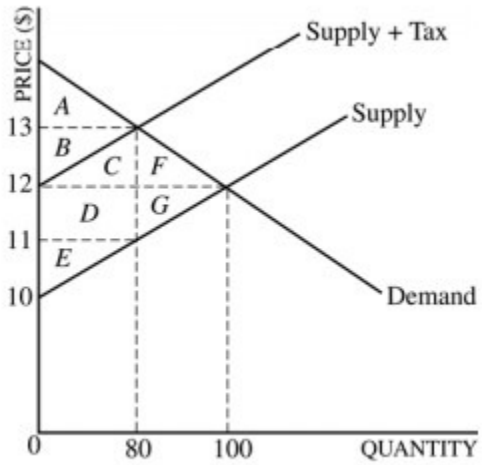


3 Points

- One point is earned for identifying P = $12 and Q = 100 units.
- One point is earned for identifying the consumer surplus before the tax, A + B + C + F.
- One point is earned for identifying the producer surplus before the tax, D + E + G or $100.

0	1	2	3 ✓

The student earns all of the following points:



- One point is earned for identifying P = $12 and Q = 100 units.

2.8 government intervention FRQ

- One point is earned for identifying the consumer surplus before the tax, A + B + C + F.
- One point is earned for identifying the producer surplus before the tax, D + E + G or $100.

Part B

2 Points

- One point is earned for stating that the price paid by the buyers does not rise by the full amount of the tax.
- One point is earned for a correct explanation: P increases by $1 and the tax is $2 per unit; or a correct elasticity explanation, such as S is not perfectly elastic, or D is not perfectly inelastic.

0	1	2 ✓

The student earns all of the following points:

- One point is earned for stating that the price paid by the buyers does not rise by the full amount of the tax.
- One point is earned for a correct explanation: P increases by $1 and the tax is $2 per unit; or a correct elasticity explanation, such as S is not perfectly elastic, or D is not perfectly inelastic.

Part C

4 Points

- One point is earned for identifying the net price received by the sellers, $11.
- One point is earned for identifying the tax revenue, B + C + D or $160.
- One point is earned for identifying consumer surplus, A.
- One point is earned for identifying the deadweight loss, F + G or $20.

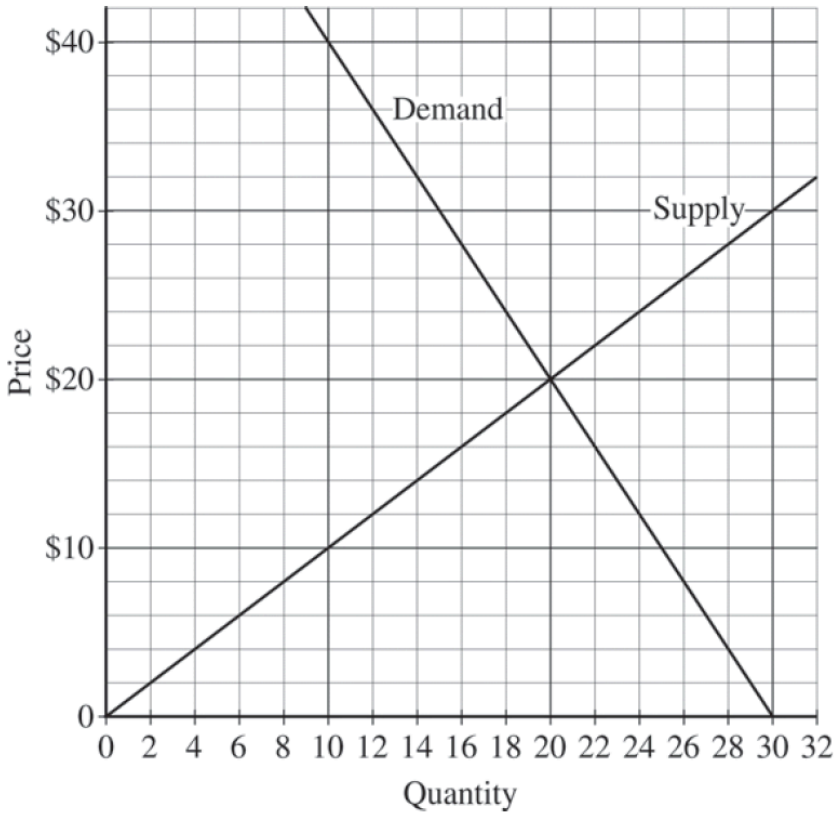
0	1	2	3	4 ✓

The student earns all of the following points:

- One point is earned for identifying the net price received by the sellers, $11.
- One point is earned for identifying the tax revenue, B + C + D or $160.
- One point is earned for identifying consumer surplus, A.
- One point is earned for identifying the deadweight loss, F + G or $20.

2.8 government intervention FRQ

The graph below shows the market for widgets. The government is considering intervening in this market.



a. Calculate the total producer surplus at the market equilibrium price and quantity. Show your work.

b. If the government imposes a price floor at $16, is there a shortage, a surplus, or neither? Explain.

c. If instead the government imposes a price ceiling at $12, is there a shortage, a surplus, or neither? Explain.

d. If instead the government restricts the market output to 10 units, calculate the deadweight loss. Show your work.

e. Assume the price decreases from $20 to $12.

   i. Calculate the price elasticity of demand. Show your work.

   ii. In this price range, is demand perfectly elastic, relatively elastic, unit elastic, relatively inelastic, or perfectly inelastic?

9. Respond to all parts of the question.

Part A

1 point:

2.8 government intervention FRQ

- One point is earned for calculating the total producer surplus as (1/2 × 20 × 20) = $200.

0	1 ✓

Student receives 1/1 point(s)

1 point:

- One point is earned for calculating the total producer surplus as (1/2 × 20 × 20) = $200.

Part B

1 point:

- One point is earned for stating that imposing a price floor at $16 is ineffective and will not create a surplus or a shortage in the market because it is set below the equilibrium price, or because it is not binding.

0	1 ✓

Student receives 1/1 point(s)

1 point:

- One point is earned for stating that imposing a price floor at $16 is ineffective and will not create a surplus or a shortage in the market because it is set below the equilibrium price, or because it is not binding.

Part C

1 point:

- One point is earned for stating that imposing a price ceiling at $12 will create a shortage because quantity demanded is greater than quantity supplied, or because the price ceiling is binding.

0	1 ✓

Student receives 1/1 point(s)

1 point:

- One point is earned for stating that imposing a price ceiling at $12 will create a shortage because quantity demanded is greater than quantity supplied, or because the price ceiling is binding.

2.8 government intervention FRQ

Part D

1 point:

- One point is earned for calculating the deadweight loss as $150 and for showing:

  (1/2 × 30 × 10)

  or

  (1/2 × 10 × 10) + (1/2 × 20 × 10)

  or

  $50 + $100

0	1 ✓

Student receives 1/1 point(s)

1 point:

- One point is earned for calculating the deadweight loss as $150 and for showing:

  (1/2 × 30 × 10)

  or

  (1/2 × 10 × 10) + (1/2 × 20 × 10)

  or

  $50 + $100

Part E

2 points:

- One point is earned for calculating the price elasticity of demand as [(24−20)/20 / (12−20)/20] = −0.5, or for correctly using the midpoint formula.
- One point is earned for stating that in this price range the demand is relatively inelastic.

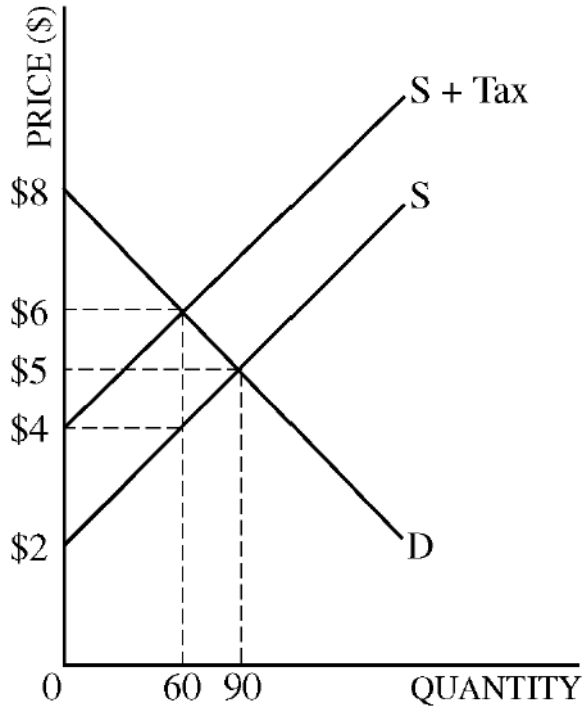
0	1	2 ✓

2.8 government intervention FRQ

Student receives 2/2 point(s)

2 points:

- One point is earned for calculating the price elasticity of demand as [(24−20)/20 / (12−20)/20] = −0.5, or for correctly using the midpoint formula.
- One point is earned for stating that in this price range the demand is relatively inelastic.



The graph above illustrates the market for calculators. S denotes the current supply curve, and D denotes the demand curve.

a. Calculate the producer surplus before the tax.

b. Now assume a per-unit tax of $2 is imposed whose impact is shown in the graph above.

   i. Calculate the amount of tax revenue.

   ii. What is the after-tax price that the sellers now keep?

   iii. Calculate the producer surplus after the tax.

c. Is the demand price elastic, inelastic, or unit elastic between the prices of $5 and $6 ? Explain.

d. Assuming no externalities, how does the tax affect allocative efficiency? Explain.

2.8 government intervention FRQ

10. Respond to all parts of the question.

Part A

1 point:

- One point is earned for the correct calculation of the producer surplus: (1/2) × $3 × 90 = $135.

0	1 ✓

Student receives 1/1 point(s)

1 point:

- One point is earned for the correct calculation of the producer surplus: (1/2) × $3 × 90 = $135.

Part B

3 points:

- One point is earned for the correct calculation of the amount of tax revenue: $2 × 60 = $120.
- One point is earned for the correct calculation of the after-tax price received by sellers: $4.
- One point is earned for the correct calculation of the producer surplus: (1/2) × $2 × 60 = $60.

0	1	2	3 ✓

Student receives 3/3 point(s)

3 points:

- One point is earned for the correct calculation of the amount of tax revenue: $2 × 60 = $120.
- One point is earned for the correct calculation of the after-tax price received by sellers: $4.
- One point is earned for the correct calculation of the producer surplus: (1/2) × $2 × 60 = $60.

Part C

1 point:

- One point is earned for concluding that the demand price is elastic AND showing the correct calculation of the elasticity coefficient using endpoint or midpoint method, or the correct calculation using the total revenue

2.8 government intervention FRQ

formula.

0	1 ✓

Student receives 1/1 point(s)

1 point:

- One point is earned for concluding that the demand price is elastic AND showing the correct calculation of the elasticity coefficient using endpoint or midpoint method, or the correct calculation using the total revenue formula.

Part D

1 point:

- One point is earned for concluding that, owing to the tax, the market is no longer allocatively efficient AND that total surplus decreases or the tax creates a dead-weight loss.

0	1 ✓

Student receives 1/1 point(s)

1 point:

- One point is earned for concluding that, owing to the tax, the market is no longer allocatively efficient AND that total surplus decreases or the tax creates a dead-weight loss.

2.8 government intervention FRQ

Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.

Assume the market for good Y is in equilibrium.

(a) Draw a correctly labeled demand and supply graph for good Y. Label the equilibrium price P_e and the equilibrium quantity Q_e.

(b) Assume the government imposes a per-unit tax on good Y. On your graph in part (a), show each of the following after the tax has been implemented.

(i) The equilibrium price labeled P_N and the equilibrium quantity labeled Q_N

(ii) The area representing the change in consumer surplus, shaded completely

(c) Will the price paid by consumers increase by the same amount as the tax? Explain.

(d) Will the loss in consumer and producer surplus be greater than, less than, or equal to the tax revenue collected by the government? Explain.

11. Respond to all parts of the question.

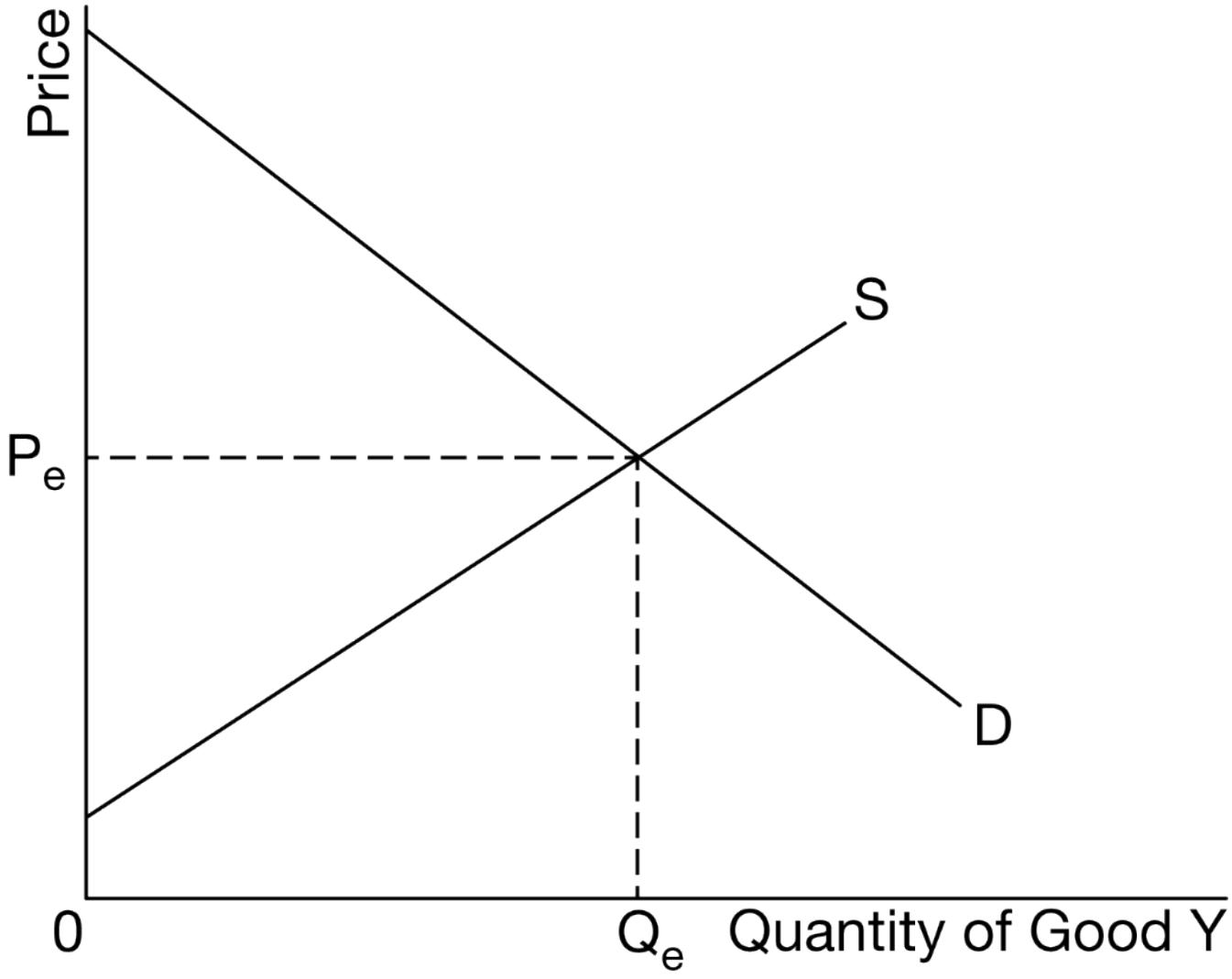
Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The response does show a correctly labeled graph of the market for good Y with the equilibrium price labeled P_e and the equilibrium quantity labeled Q_e .

2.8 government intervention FRQ



Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

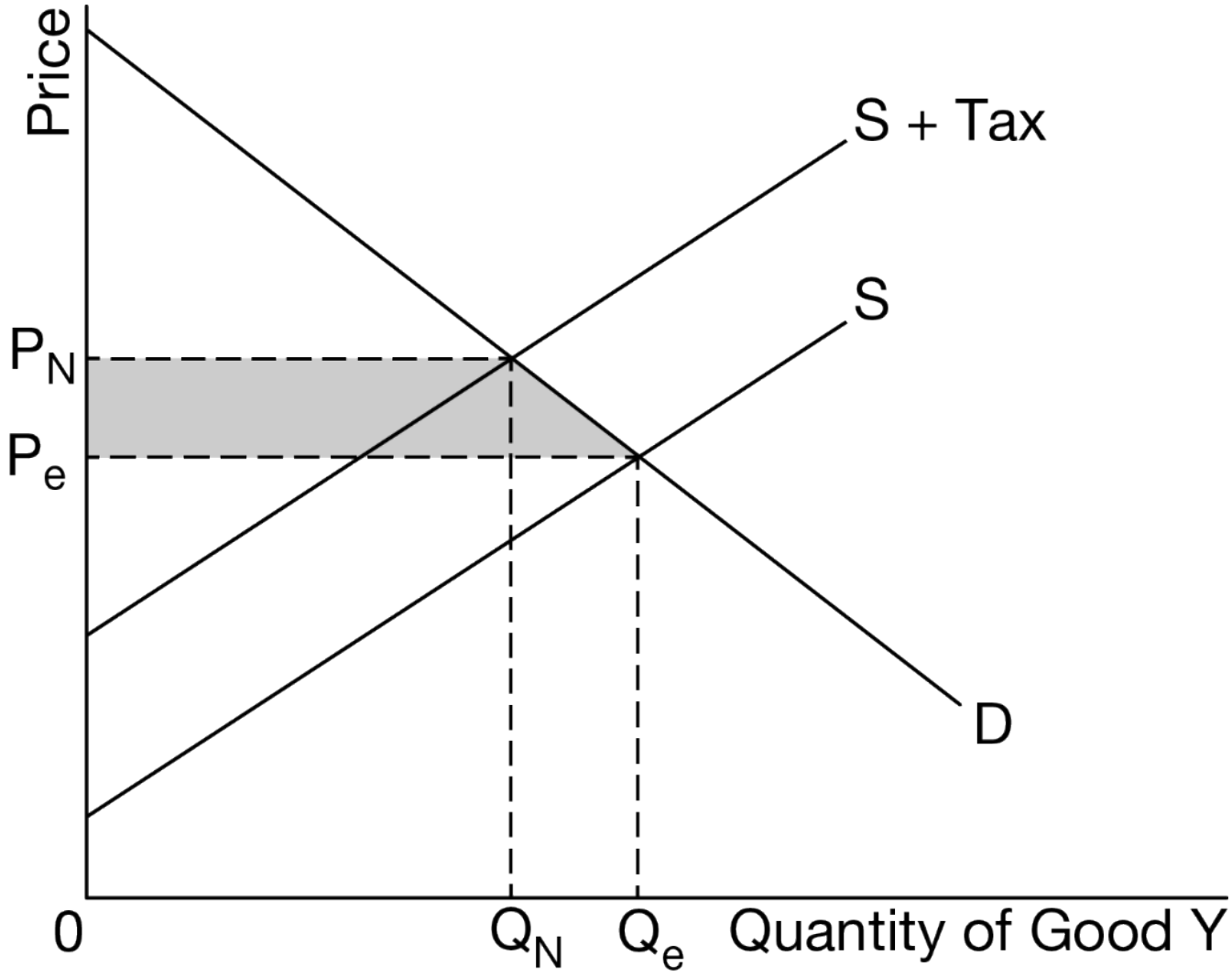
0	1	2 ✓

The response accurately includes both of the criteria below.

- ☐ The new equilibrium price labeled P_N and the new equilibrium quantity labeled Q_N
- ☐ The area representing the change in consumer surplus, shaded completely

Solution

2.8 government intervention FRQ



Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must both state that the price will increase by less than the amount of the tax and explain why in order to receive the point.

0	1 ✓

The response states that the price will increase by less than the amount of the tax and explains that, unless demand is perfectly inelastic, producers will not be able to raise price by the amount of the tax.

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2.8 government intervention FRQ

The response must both state that the loss in consumer surplus and producer surplus will be greater than the tax revenue collected by the government and explain why in order to receive the point.

0	1 ✓

The response states that the loss in consumer and producer surplus will be greater than the tax revenue collected by the government and explains that the tax reduces the quantity exchanged in the market relative to the unregulated market equilibrium quantity, resulting in a deadweight loss which is equal to the decrease in the total economic surplus.

The market for apartments in the town of Westville is currently in equilibrium, and consumers are renting 700 apartments at a price of $1,000 per apartment.

a. Assuming that the supply of apartments is perfectly inelastic, draw a correctly labeled graph for the apartment market and label the equilibrium price and quantity using the values above.

b. Now assume the government imposes a price (rent) ceiling of $800 for an apartment. On your graph in part (a), show the following.

   i. The price ceiling at $800

   ii. The quantity demanded, labeled as Q_d

   iii. The quantity supplied, labeled as Q_s

c. On your graph, shade completely the area representing the consumer surplus after the imposition of the price ceiling.

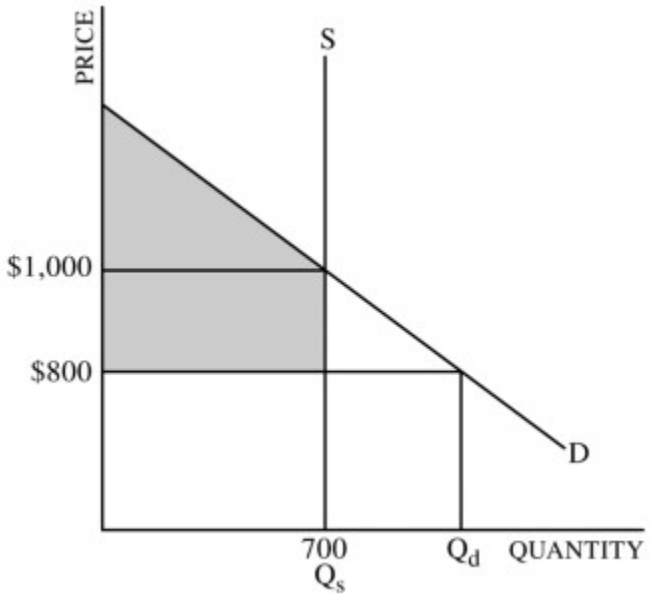
d. Calculate the dollar value of the producer surplus after the imposition of the price ceiling. Show your work.

e. Explain why the imposition of the price ceiling does not result in a deadweight loss.

12. Respond to all parts of the question.

Part A

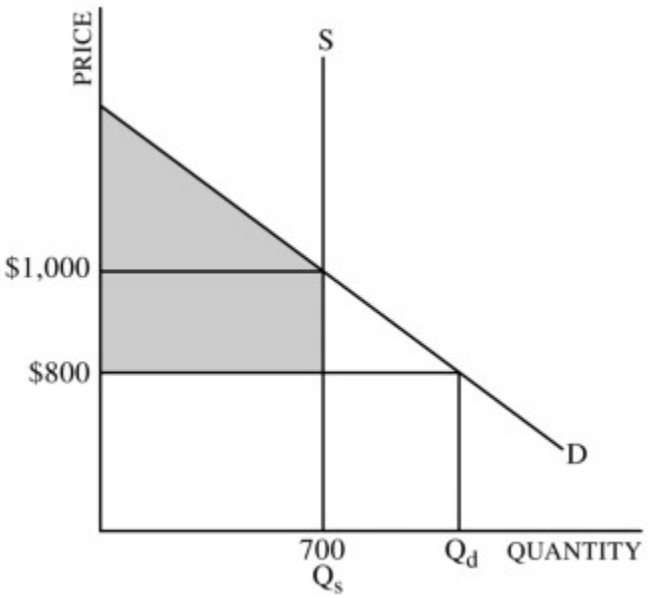
2.8 government intervention FRQ



One point is earned for drawing a graph with correctly labeled axes, supply, demand, and identifying equilibrium price and quantity as $1000 and 700.

0	1 ✓

The student earns all of the following points:

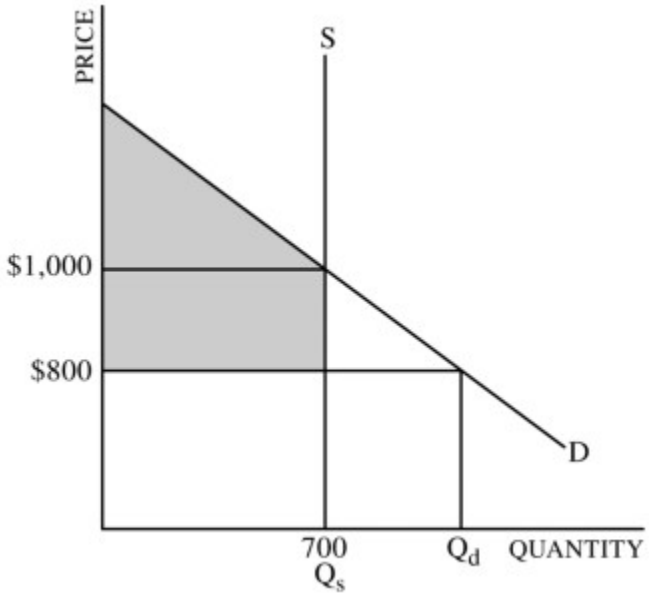


One point is earned for drawing a graph with correctly labeled axes, supply, demand, and identifying equilibrium price

2.8 government intervention FRQ

and quantity as $1000 and 700.

Part B



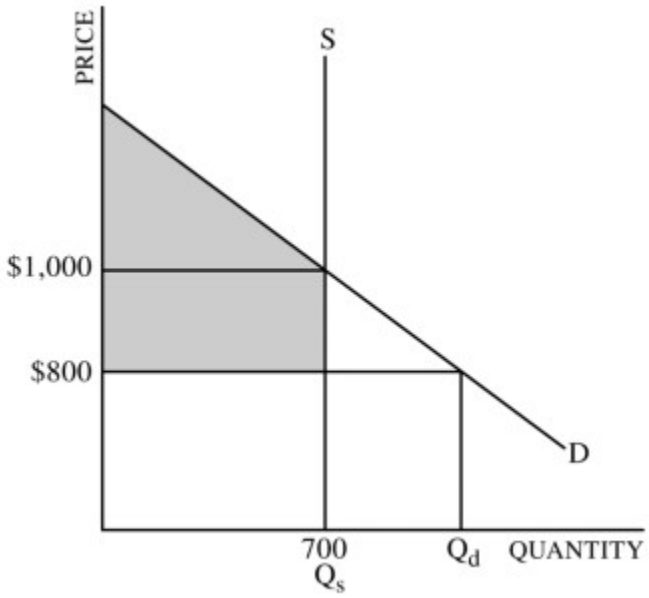
3 Points

- One point is earned for showing the price ceiling of $800 below $1000.
- One point is earned for showing Qd derived from the demand curve at $800.
- One point is earned for showing Qs derived from the supply curve at $800.

0	1	2	3 ✓

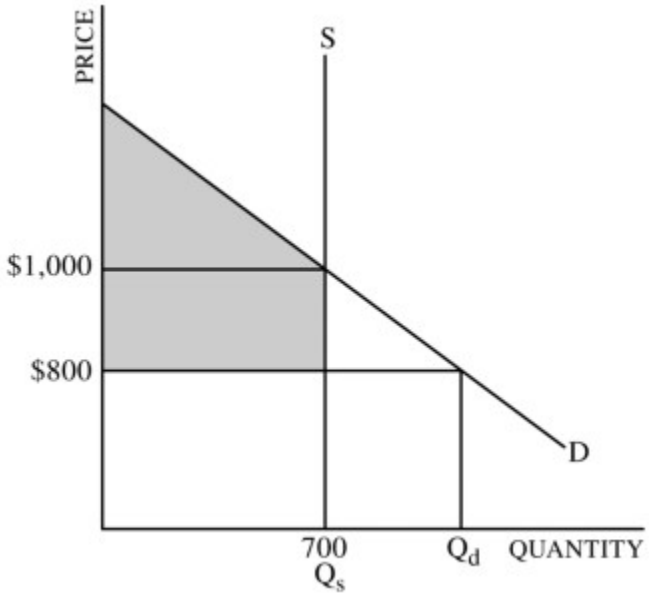
The student earns all of the following points:

2.8 government intervention FRQ



- One point is earned for showing the price ceiling of $800 below $1000.
- One point is earned for showing Qd derived from the demand curve at $800.
- One point is earned for showing Qs derived from the supply curve at $800.

Part C



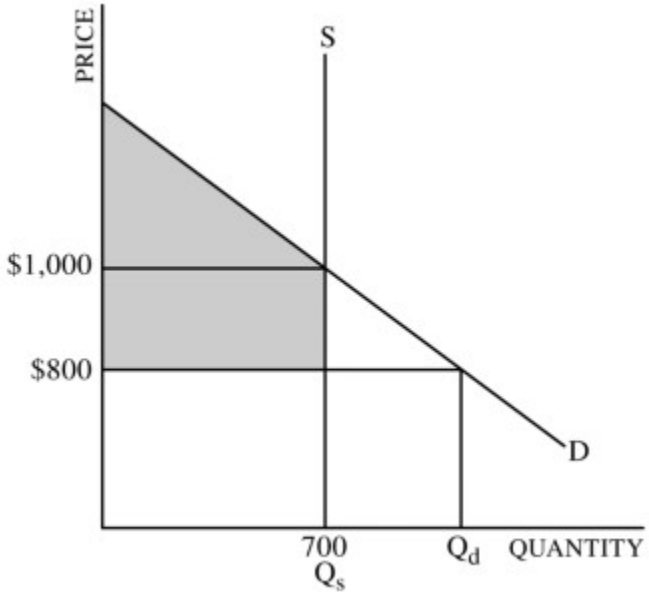
One point is earned for showing the area representing the consumer surplus shaded completely.

Note: If the supply curve from part (a) is not perfectly inelastic, no points can be awarded in parts (d) and (e).

2.8 government intervention FRQ

0	1 ✓

The student earns all of the following points:



One point is earned for showing the area representing the consumer surplus shaded completely.

Part D

One point is earned for calculating the producer surplus: $800×700 = $560,000.

0	1 ✓

The student earns all of the following points:

- One point is earned for calculating the producer surplus: $800×700 = $560,000.

Part E

One point is earned for the explanation that the quantity supplied does not change after the imposition of the price ceiling.

2.8 government intervention FRQ

0	1 ✓

The student earns all of the following points:

- One point is earned for the explanation that the quantity supplied does not change after the imposition of the price ceiling.

---

13. Respond to all parts of the question.

Directions: You are advised to spend the first 10 minutes reading all of the questions and planning your answers. You will then have 50 minutes to answer all three of the following questions. You may begin writing your responses before the reading period is over. It is suggested that you spend approximately half your time on the first question and divide the remaining time equally between the next two questions.Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. Use a pen with black or dark blue ink.

Assume milk is produced and sold in a perfectly competitive market. To help milk producers, the government imposes an effective price floor on milk. Assume milk producers continue to sell milk only to households after the imposition of the price floor.

1. Draw a correctly labeled graph for the milk market and show the price floor, labeled P_F , and the quantity of milk purchased by households, labeled Q_F.

2. Is the market for milk allocatively efficient after the price floor is imposed? Explain.

3. On your graph from part (a), shade the area that represents total producer surplus after the price floor is imposed.

4. Assume the absolute value of price elasticity of demand for milk is 2 throughout the price range. Would total consumer spending on milk increase, decrease, not change, or be indeterminate after the imposition of the price floor? Explain.

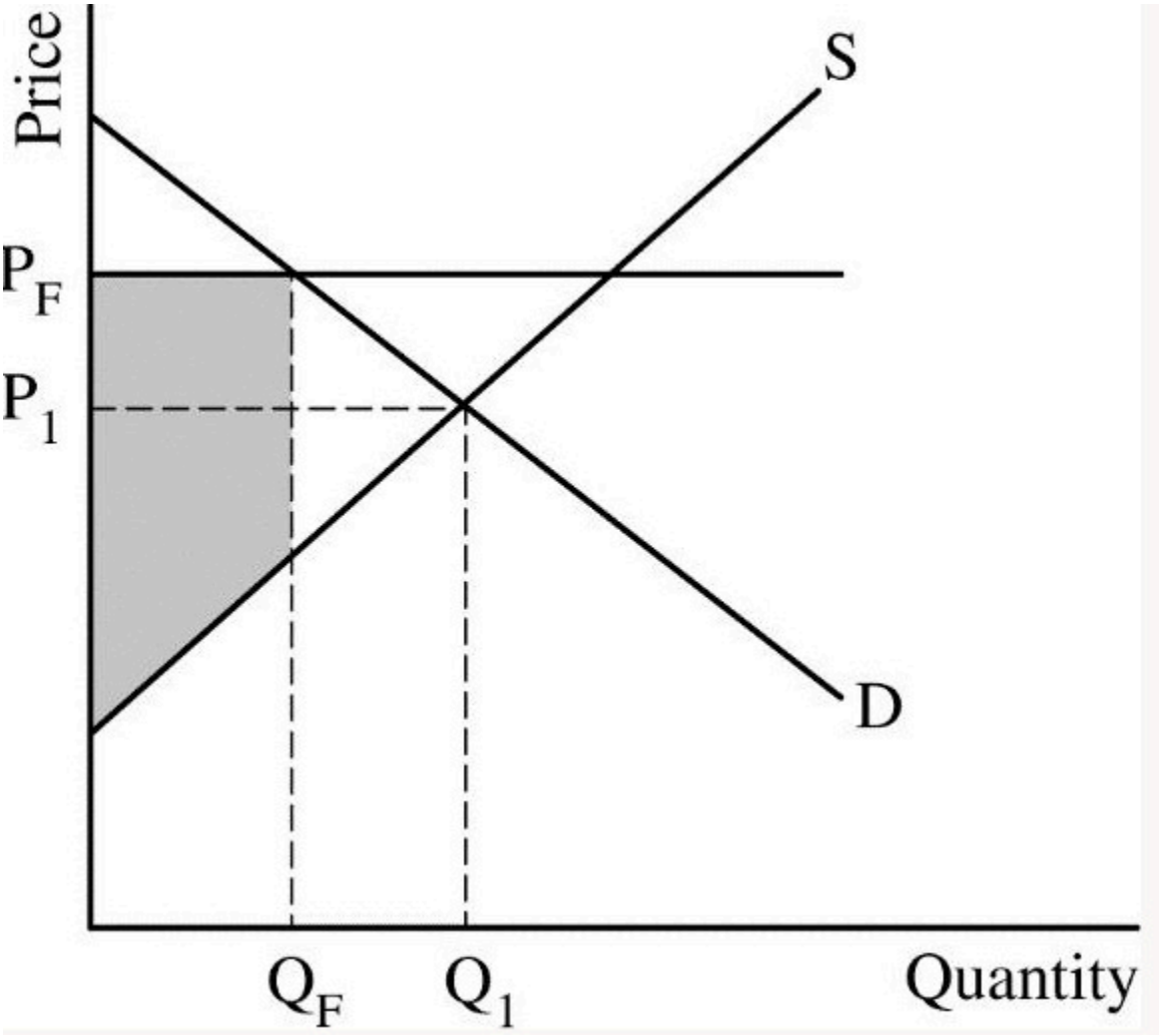
5. Assume the price floor in the milk market caused the equilibrium price of eggs, a related consumption good, to increase.

    1. Draw a correctly labeled graph of the egg market, and show the effect of the price floor in the milk market on the egg market.

    2. What economic term describes the relationship between milk and egg?

Part A

2.8 government intervention FRQ



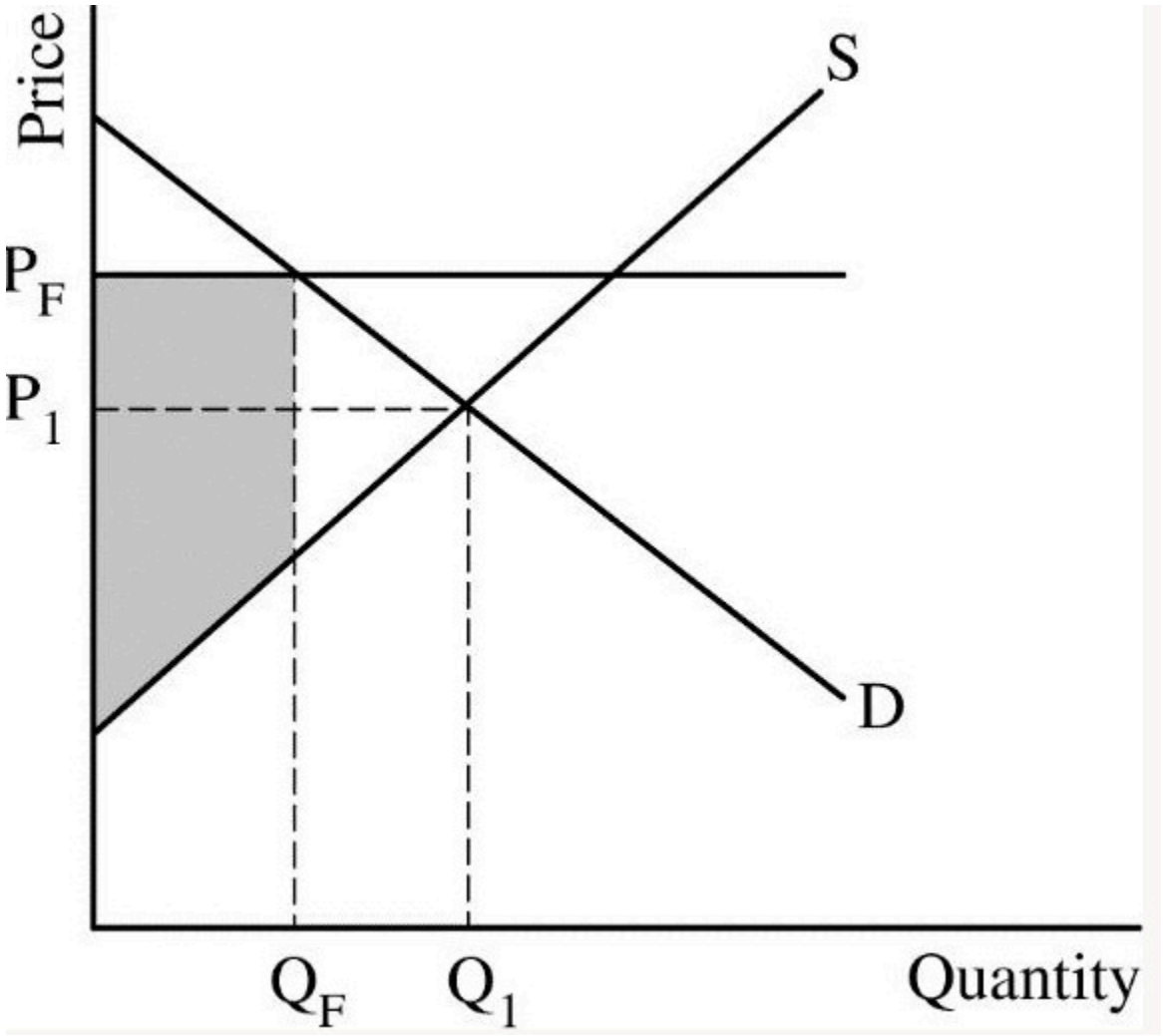
2 points:

- One point is earned for drawing a correctly labeled graph of the market for milk with an upward sloping supply curve and a downward sloping demand curve.
- One point is earned for drawing an effective price floor, labeled PF, above the equilibrium price and for identifying the new quantity of milk, labeled QF, from the demand curve.

0	1	2 ✓

Student receives 2/2 Point(s)

2.8 government intervention FRQ



2 points:

- One point is earned for drawing a correctly labeled graph of the market for milk with an upward sloping supply curve and a downward sloping demand curve.
- One point is earned for drawing an effective price floor, labeled PF, above the equilibrium price and for identifying the new quantity of milk, labeled QF, from the demand curve.

Part B

1 point:

- One point is earned for stating the market is not allocatively efficient after the price floor is imposed and for explaining that marginal benefit is greater than marginal cost (MB > MC).

0	1 ✓

2.8 government intervention FRQ

Student receives 1/1 Point(s)

1 point:

- One point is earned for stating the market is not allocatively efficient after the price floor is imposed and for explaining that marginal benefit is greater than marginal cost (MB > MC).

Part C

1 point:

- One point is earned for completely shading the area representing the producer surplus.

0	1 ✓

Student receives 1/1 Point(s)

1 point:

- One point is earned for completely shading the area representing the producer surplus.

Part D

1 point:

- One point is earned for stating that total consumer spending on milk will decrease, and for explaining the demand is elastic or that the percentage increase in the price of milk will be less than the percentage decrease in the quantity demanded.

0	1 ✓

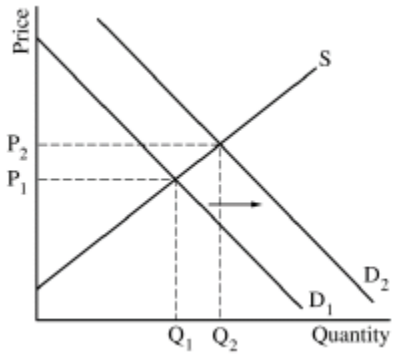
Student receives 1/1 Point(s)

1 point:

- One point is earned for stating that total consumer spending on milk will decrease, and for explaining the demand is elastic or that the percentage increase in the price of milk will be less than the percentage decrease in the quantity demanded.

Part E

2.8 government intervention FRQ

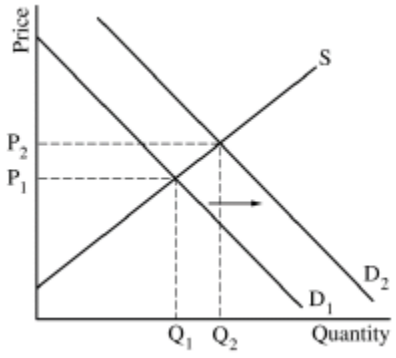


2 points

- One point is earned for drawing a correctly labeled graph for the egg market with an upward sloping supply curve and a downward sloping demand curve, and for showing a rightward shift in the demand curve for eggs.
- One point is earned for stating that the two goods are substitutes.

0	1	2 ✓

Student receives 2/2 Point(s)



2 points

- One point is earned for drawing a correctly labeled graph for the egg market with an upward sloping supply curve and a downward sloping demand curve, and for showing a rightward shift in the demand curve for eggs.
- One point is earned for stating that the two goods are substitutes.

2.8 government intervention FRQ

a. The table below gives the quantity of good X demanded and supplied at various prices.

Price (dollars)	Quantity Demanded (units)	Quantity Supplied (units)
30	1	3
20	3	3
10	4	3

i. Is the demand for good X relatively elastic, relatively inelastic, unit elastic, perfectly elastic, or perfectly inelastic when the price decreases from $30 to $20 ? Explain.

ii. Is the supply of good X relatively elastic, relatively inelastic, unit elastic, perfectly elastic, or perfectly inelastic when the price decreases from $30 to $20 ? Explain.

iii. If a per-unit tax is imposed on good X, how is the burden of the tax distributed between the buyers and sellers of good X?

b. Assume that the income elasticity of demand for good Y is –2. Using a correctly labeled graph of the market for good Y, show the effect of a significant increase in income on the equilibrium price of good Y in the short run.

14. Respond to all parts of the question.

Part A

3 points:

- One point is earned for stating that the demand for good X is relatively elastic, because the elasticity coefficient > 1 OR because total revenue rises as price decreases from $30 to $20.
- One point is earned for stating that supply is perfectly inelastic, because there is no change in the quantity supplied as the price changes OR because the supply elasticity is zero.
- One point is earned for stating that all of the burden of a per-unit tax falls on sellers.

0	1	2	3 ✓

Student receives 3/3 point(s)

3 points:

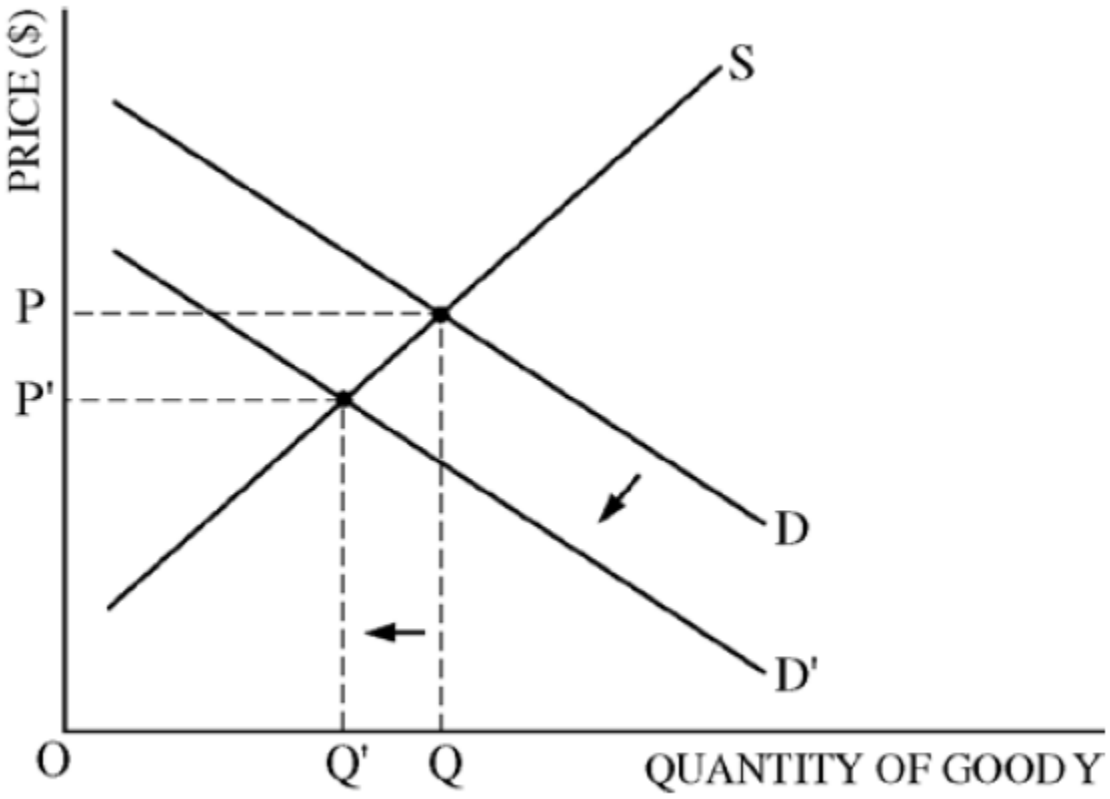
- One point is earned for stating that the demand for good X is relatively elastic, because the elasticity coefficient > 1 OR because total revenue rises as price decreases from $30 to $20.
- One point is earned for stating that supply is perfectly inelastic, because there is no change in the quantity

2.8 government intervention FRQ

supplied as the price changes OR because the supply elasticity is zero.

- One point is earned for stating that all of the burden of a per-unit tax falls on sellers.

Part B



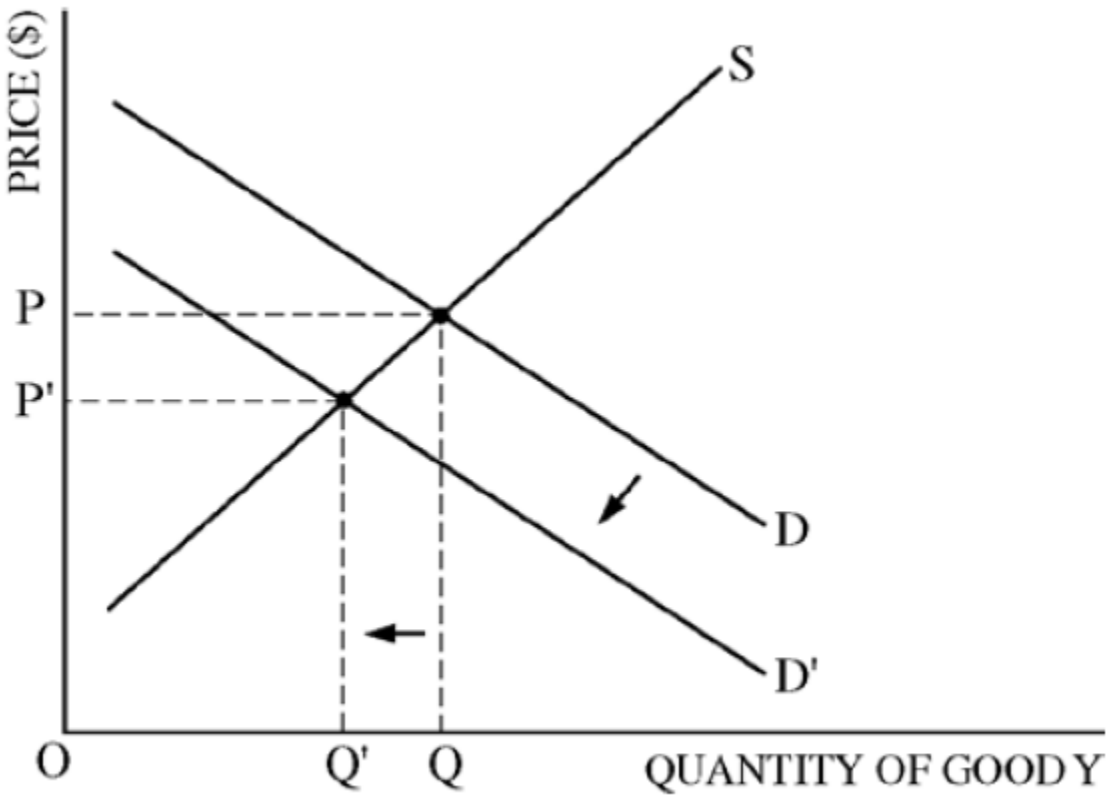
2 points:

- One point is earned for a correctly labeled graph of supply and demand.
- One point is earned for showing a leftward shift of the demand curve and a decrease in the equilibrium price.

0	1	2 ✓

Student receives 2/2 point(s)

2.8 government intervention FRQ

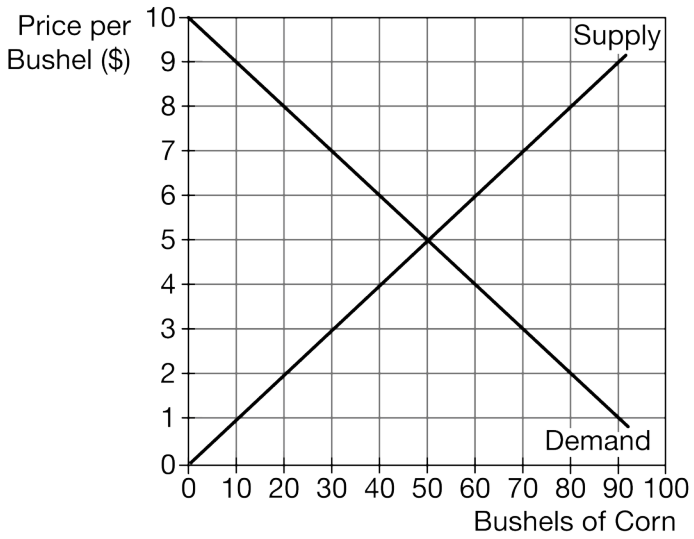


2 points:

- One point is earned for a correctly labeled graph of supply and demand.
- One point is earned for showing a leftward shift of the demand curve and a decrease in the equilibrium price.

2.8 government intervention FRQ

Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.



The diagram above shows the market for corn in the country of Microland. Corn is produced and sold in a constant-cost, perfectly competitive market.

(a) Calculate the total revenue earned by corn farmers at the market equilibrium price. Show your work.

(b) In an attempt to assist corn farmers in Microland, the government sets a $7 price floor on corn.

(i) How many bushels of corn will be exchanged at the price floor?
(ii) Calculate the deadweight loss associated with the price floor. Show your work.
(iii) Assume the government agrees to buy the unsold quantity at $7. Calculate the producer surplus. Show your work.
(iv) Assume the price floor and the government buying program remain in effect. In addition, assume the demand for corn does not change. In the long run, will the quantity of corn purchased by the government increase, decrease, or remain the same? Explain.

15. Respond to all parts of the question:

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must show appropriate work in order to receive the point.

0	1 ✓

2.8 government intervention FRQ

The response calculates the total revenue at the market price as $\$250$ and shows the work.

$$\text{Total Revenue} = (\text{P} \times \text{Q}) = (\$5 \times 50) = \$250$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must show appropriate work in order to receive the point.

0	1	2	3	4 ✓

The response accurately includes the four criteria below.

- ☐ States that the quantity exchanged at the price floor will be 30 bushels.
- ☐ Calculates the deadweight loss as $\$40$ and shows the work.
  $$\text{Deadweight loss} = \frac{(\$7 – \$3) \times (50 – 30)}{2} = \frac{\$80}{2} = \$40$$
- ☐ Calculates the producer surplus as $\$245$ and shows the work.
  $$\text{Producer surplus} = \frac{(\$7 – \$0) \times (70 – \$0)}{2} = \frac{\$490}{2} = \$245$$
- ☐ States that the quantity of corn purchased by the government will increase and explains that this happens because the price floor leads to short-run economic profits for typical firms, which incentivizes new firms to enter the market, leading to an increase in the market supply of corn and/or because the market supply becomes more elastic in the long run.